

Paragraph-Based Keystroke Verification: A Prototype Toward Continuous Authentication

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Current implementation: Verification after a user types a fixed paragraph.

Future goal: Unobtrusive, continuous verification during normal typing.

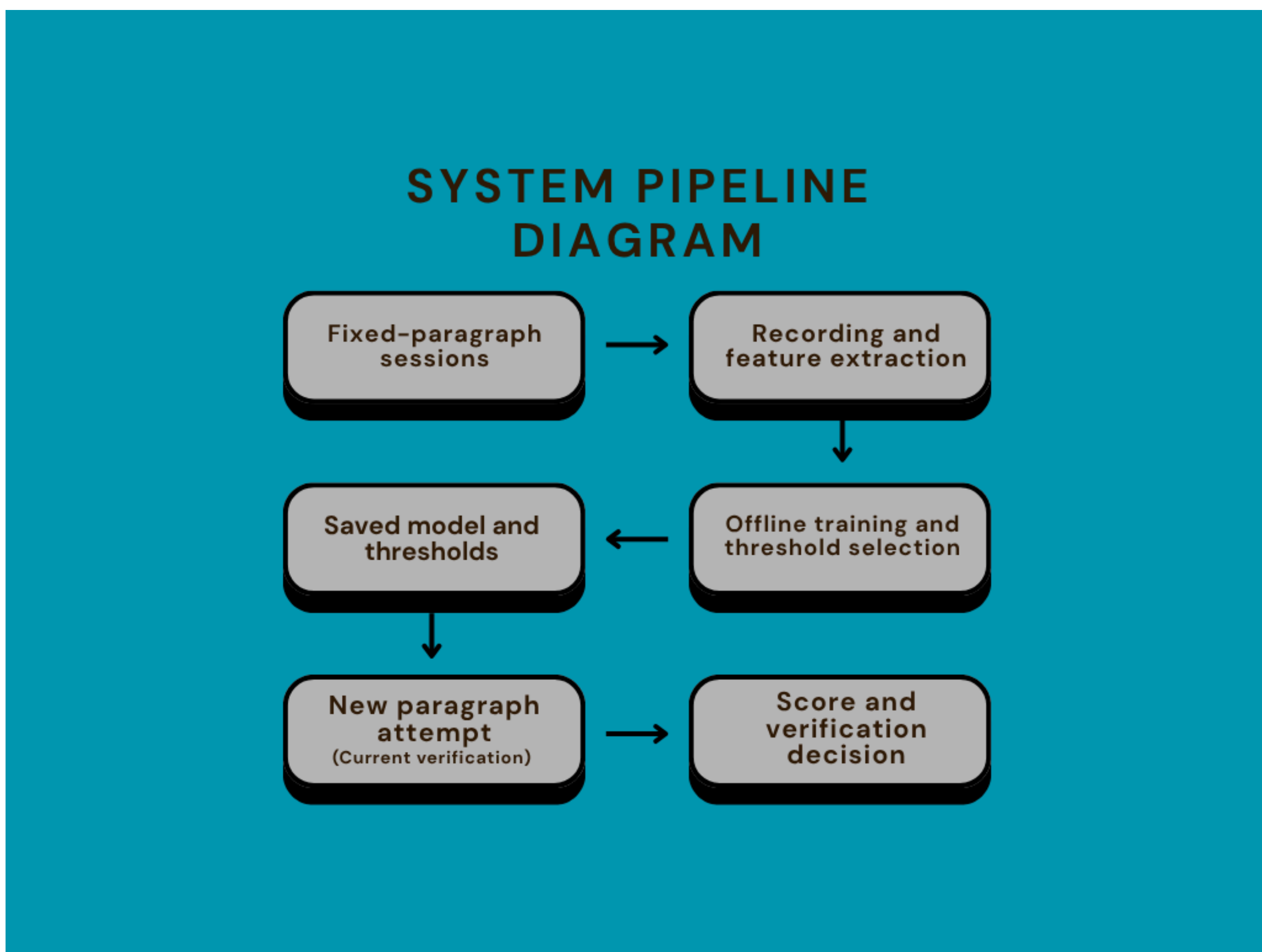
1) Problem and Motivation

- A login check does not establish who continues using an active session.
- Keystroke timing may provide an additional signal about the person typing.
- This prototype studies that signal through a fixed paragraph, as a first step toward continuous authentication.

2) What the Prototype Does Today

- Records key-down and key-up timing while a participant types a fixed paragraph.
- Converts recorded sessions into features for offline preprocessing and model training.
- Uses a previously trained model and stored thresholds to make one verification decision after a completed paragraph.

3) System pipeline diagram



4) Method

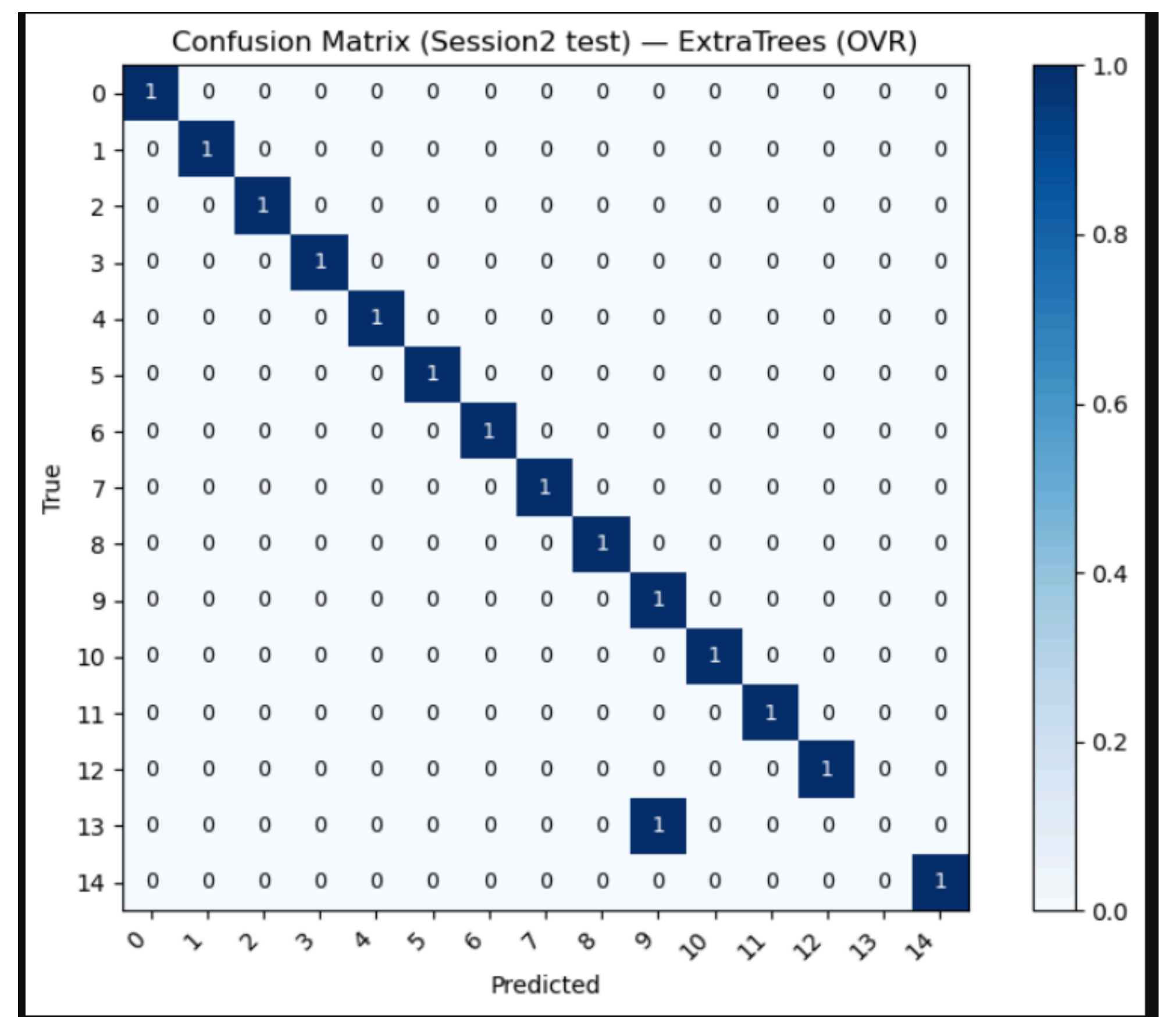
- The browser records key-down and key-up timestamps during the fixed-paragraph task.
- Timing features include dwell time (how long a key is held) and flight time (timing between keys).
- The active Random Forest model uses 501 input features: 483 key-timing features and 18 typing, error, speed, and text-length measures.
- Recorded sessions are aggregated and preprocessed. An offline-trained classifier scores a new attempt against the claimed identity; a stored threshold contributes to the accept/reject decision.

5) Outputs

- The typing task can display speed, accuracy, and timing summaries.
- The paragraph verification page shows the claimed identity, model score, threshold, and accept/reject result after the attempt is complete.
- Recorded data can be exported for offline analysis and training.

6) Evaluation and Scope

- Offline paragraph identification experiments explored 16 classifier choices.
- In an exploratory, two-fold, session-grouped identification comparison on 30 paragraph records from 15 participants, ExtraTrees achieved 93.33% mean accuracy and Random Forest achieved 90.00%.
- The displayed confusion matrix illustrates an ExtraTrees identification experiment.
- The paragraph verification prototype currently loads a Random Forest model and stored thresholds. Identification accuracy does not measure its false acceptance or false rejection rates.
- Reliable verification evaluation requires more independent enrollment, threshold-calibration, and test attempts.



ExtraTrees identification confusion matrix; separate from the Random Forest model used by the verification page.

Scan to view the public source code, workflow, and sample data.

